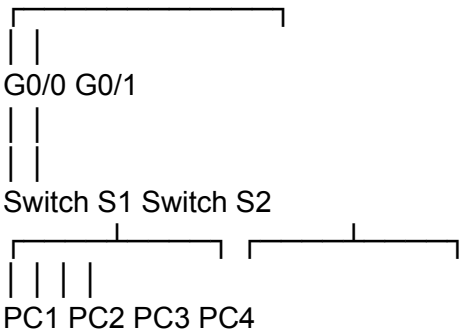
PRAVINTHEKAA R.L 251901128

Ex.No.10 Design a simple topology and configure with one router, two

switches and PCs using Cisco Packet Tracer

DATE:
AIM:
To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.
PROCEDURE:

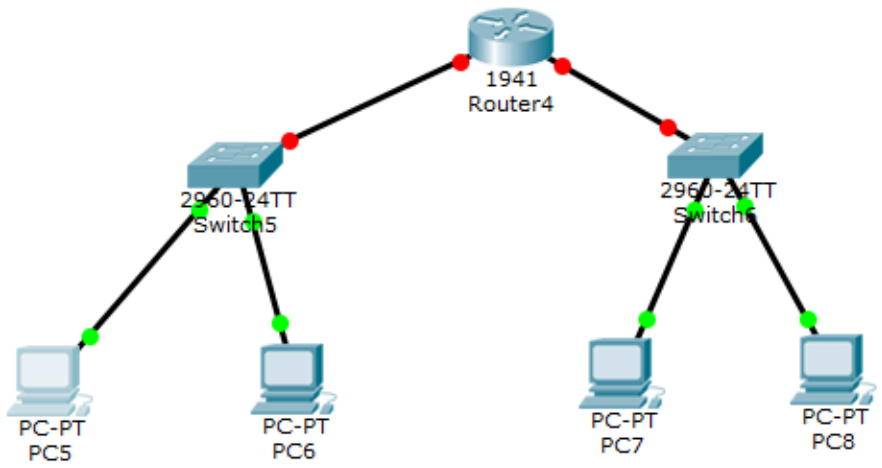
1. Topology
Router R1



2. Devices to place
In Cisco Packet Tracer, place:
1 Router – Cisco 1941
2 Switches – Cisco 2960
4 PCs
Use Copper Straight-Through cables.
3. IP Addressing
Device Interface IP Address Subnet Mask Gateway
R1 G0/0 192.168.1.1 255.255.255.0 —
R1 G0/1 192.168.2.1 255.255.255.0 —
PC1 NIC 192.168.1.10 255.255.255.0 192.168.1.1
PC2 NIC 192.168.1.11 255.255.255.0 192.168.1.1
PC3 NIC 192.168.2.10 255.255.255.0 192.168.2.1
PC4 NIC 192.168.2.11 255.255.255.0 192.168.2.1
So we have two different networks:

Network 1: 192.168.1.0/24
PC1 ─┐
PC2 ─┤
S1 ─── R1 G0/0 = 192.168.1.1
Network 2: 192.168.2.0/24

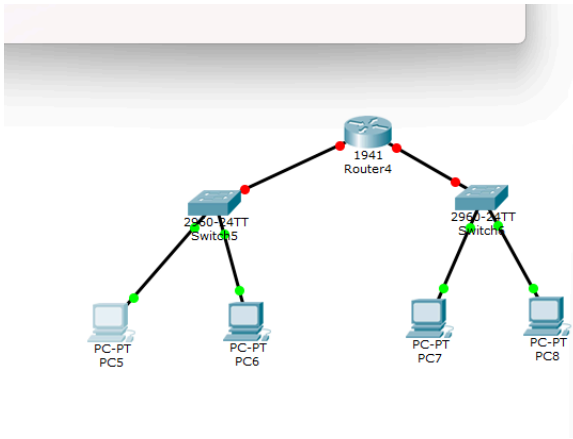
PC3 ─┐
PC4 ─┤
S2 ─── R1 G0/1 = 192.168.2.1
4. Configure Router R1
Click R1 → CLI and enter:
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
end
write memory
The important command is:
no shutdown
It activates the router interface.
Check the interfaces using:
show ip interface brief
You should see the interfaces as up/up.

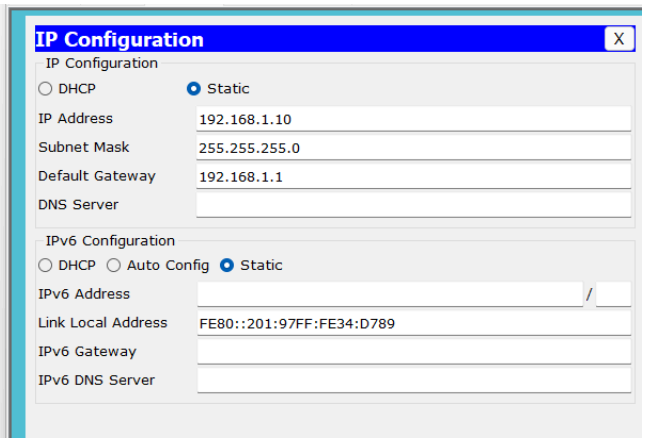


5. Configure PC1
Go to:
PC1 → Desktop → IP Configuration
Enter:
IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

6. Configure PC2
IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

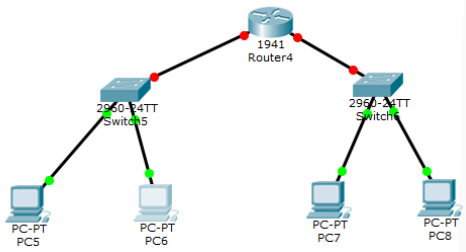
7. Configure PC3
IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

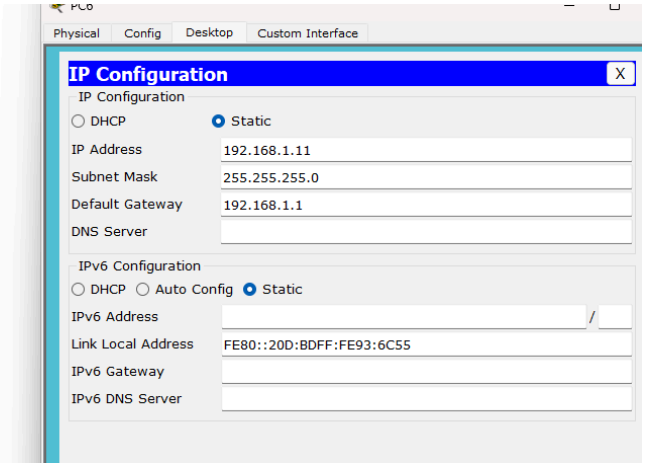




8. Configure PC4
IP Address: 192.168.2.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

9. Test the network
From PC1 → Desktop → Command Prompt, type:
ping 192.168.1.11
This tests PC1 → PC2.
Then:





ping 192.168.2.10
This tests PC1 → Router → Switch 2 → PC3.
You should receive:
Reply from 192.168.2.10
Reply from 192.168.2.10
Reply from 192.168.2.10
Reply from 192.168.2.10

SAMPLE CONFIGURATION:

RESULT:
THUS TO Design a simple topology and configure with one router, two

switches and PCs using Cisco Packet Tracer COMPLETED.